



UNIVERSIDADE D
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REFLEXIVE ESSAY

SOCIAL AND COLLABORATIVE COMPUTING

Title: Tell me Something

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Introduction

We live in era where digital interactions increasingly replace physical interactions. Our project “Tell Me Something” comes in reply to Byung-Chul Han’s critique of the “non-things” that dominate our virtual experiences. Byung-Chul Han warns us about the fragility of the collective memory and the superficiality of digital interactions in “Non-Things”, in this work he analyzes how people’s obsession with information and data in general has affected and led to the loss of tangible and meaningful connections. Our project “Tell Me Something” tries to avoid that and focuses on creating a virtual environment that restores the sense of belonging in a community and collective construction that defines real-world communities.

In this essay, my goal is to critically analyze the design proposal and examine how it can support cooperative work dynamics, foster shared consciousness, function as a shared information space, engage the participation of users, allow a variety of temporal and spatial investigations, efficiently make use of technical modules and last but not least allow malleability and appropriation. Based on the frameworks presented in our course ,Collaborative and Social Computing, I will show how our platform bridges theory and practice to create meaningful digital interactions that maintain and strengthen tacit knowledge exchange.

Design Proposal: Ethnographic Foundations

My ethnographic study revealed a basic human need for situated knowledge exchange, the kind that naturally takes place in common physical locations like community centers, cafes and workshops. Participants usually described the importance of observation ,imitation and practice in a community can set the best way to acquire tacit knowledge, which is the type of knowledge that is hard to define or even express. This is connected with the Legitimate Peripheral Participation(LPP) theory of the Lave and Wenger(1991), in which is explained how newcomers learn by progressively increasing their participation in community practices.

Through the design of “Tell Me Something” all of these insights transform into a virtual environment which is structures as a procedurally generated 2D world with different buildings each one representing activities(cooking, arts, repairs ,language learning, grandma’s house, etc). Each one of the buildings contains rooms for specific sub-activities, resulting to what Oldenburg(1989) would call “third places”. These “third places” are basically informal meeting spots where communities are formed between homes and workplaces (like town squares or parks). This concept turns digital interactions into something more concrete, like a physical space ,so the users don’t feel as disconnected or abstract.

Our platform combines asynchronous interactions and knowledge sharing(persistent artifacts that are left on the digital billboards) and also synchronous interactions and knowledge sharing(real-time videochats in the themed rooms).This dual approach focuses on Bardram's(1998) observation that cooperative work dynamics change between different temporal modes.The design of "Tell Me Something" also implements a reputation system that promotes and encourages meaningful participation which addresses Grudin's(2001) warning about the difficulties of evaluating groupware systems.

Supporting Three Levels of Cooperative Work Dynamics

Bardram's framework about coordination, cooperation and co-construction provides important information for understanding how our platform can support different levels of collaborative work:

Coordination is made simpler by the physical organization of the virtual city. The users of the platform can navigate to buildings and rooms that much more their interests, in the same way that one might visit various departments in a real community center. The interface of the platform makes these pathways easy and clear to understand, supporting what Schmidt and Bannon(1992) refer to as "The act of managing interdependencies between activities performed to achieve a goal".

Cooperation evolves in the themed created rooms where users participate in group activities. The live videochat gives the opportunity and enables the real-time problem solving and knowledge sharing. All the above create what Bardrma would consider as a cooperative environment where participants "reflect on the means of work". For example, in the "Repairs" room, users may work together to solve a mechanical problem by combining their knowledge.

Co-construction appears through the artifact system that our platform provides. After users joining a live videochat, they are able to leave behind multimedia artifacts (texts,images,videos) that basically become part of the room's persistent knowledge base.This is consistent with the level of collaboration analyzed by Bardram where participants "reflect on the object of work". At the end, all of the artifacts create what Bannon and Bodker(1997) would refer to as Common Information Space(CIS), which is a shared repository that develops through community participation.

These transitions between the three levels naturally are supported from the platform.For example, in the "Language" room, the user may first coordinate their arrival to the room,then participate in a live conversation about verbs and lastly add a tutorial video that will be included in the room's permanent artifacts. This pathway

reflects Bardram's observation that in real-world activities contexts, collaborative activities dynamically move between levels.

Promoting Shared Consciousness: From I-Awareness to We-Awareness

In our platform's design, Tenenberg, Roth and Socha's (2016) concept of moving from I-awareness to We-awareness is central. The spatial interface that makes the individual contributions visible within the virtual environment supports I-awareness or the ability to understand one's own actions in context. When the user leaves artifacts in a room, they are able to see it next to other artifacts and realize how they have personally contributed to the shared space.

There is a number of design elements created that demonstrate the we-awareness or the shared understanding of team activities:

Social intentionality : The platform uses reputation scores, activity feeds and avatar presence indicators to show social information. The users can see who is in the room, which artifacts have been added recently and which contributions are highly valued.

Recursive social inference: The platform has a "thank you" system that creates a feedback loop where users can understand the intentions of others based on how they show appreciation. The users learn how much the community appreciates their contributions when someone gives a "thank you" to them.

Intentional collaboration: Users working together to decorate rooms and buildings creates a shared sense of community and purpose. Users with high reputation can create and suggest new buildings, which then become tools for everyone, matches the idea that teamwork can thrive when people have a common goal and know their part.

An example of the We-awareness could be demonstrated inside the Art's Building, a "Knitting Techniques" room where participants can create common rules for how to interact. The new users can quickly notice these rules by how the artifacts and the space is organized and how the group communicates.

Functioning as Shared Information Space and Organizational Memory

Our platform follows Bannon and Bodker's (1997) concept of a Common Information Space as it acts like a shared space where all the users can access and understand the information even if they have different viewpoints. The billboards in each room of

the buildings work as boundary objects. They are basically artifacts flexible enough to adapt to different viewpoints while at the same time they are structured to be meaningful to all.

The artifact system addresses what Corbett(2000) names as key challenges of organizational memory: acquisition, retention, maintenance, search, and retrieval of knowledge. Artifacts are:

- Acquired through user contributions
- Retained on the digital billboards of the rooms
- Maintained through the community moderation
- Searchable through categories
- Retrievable through temporal organization and reputation ranking

In contrast with the traditional social media where the content disappears after one scroll, our platform organizes content in a way that lets the user explore ideas like a map which creates paths between related topics. For example, if a user is interested in Italian cuisine, he can easily and naturally move between recipes, history and cooking techniques, just like clicking links in an article.

In the platform, the archive feature keeps old content saved but out of the way, so things do not get too cluttered. In that way, it avoids overload like Han's worry about too much digital noise and also it saves memory like what Corbett calls "prosthetic memory" which is like an external brain that keeps the things you do not need now but might later. It is a balance to keep knowledge without drowning in it, just like Bannon and Bodker analyze in their work about the shared information spaces.

Designing for Engagement, Practice and Participatory Culture

Tell Me Something is designed to encourage real and ongoing participation not just clicks and scrolling. It is based on the participatory culture, where users do not just consume content but instead they become part of it and help create it. Our platform accomplishes this by making digital interaction feel more like a community life and hands on learning.

Engagement through real-life context: Our platform uses themed buildings and rooms (like cooking, arts, languages, etc.) to represent real life activities. This provides to the users a familiar place to go, like visiting a workshop. Based on Shove et al. (2012) this design encourages and helps people get involved through everyday practice.

Learning as part of participating: People learn best by doing. In our platform, users can walk around, talk with other users, join live events or find helpful content-artifacts. Newcomers start slowly and become more active members over

time(exactly like joining a real community), reflecting Lave and Wenger's(1991) idea of Legitimate Peripheral Participation

Encouraging active participation: In the platform, we built a reputation system and a “thank you” button to reward users' contributions. These features are helping to highlight which contributions were helpful. When users gain reputation, they unlock new abilities like the creation of a building. This allows them to be co-creators in the platform and not just simple visitors.

Small rituals build connection: Simple actions of the users like give a “thank you”, leave an artifact behind or hosting an event help the community feel alive. Roque(2020) refers to them as collective situated actions, where like small repeated actions help to form a group identity and memory.

A friendly and open space: Ethnographic findings show that many people prefer to learn relaxed and in informal environments. Based on that, Tell Me Something avoids rigid roles and encourages fluidity. Anyone can learn and share knowledge. This matches Oldenburg's(1989) idea of third places(like parks or cafes) where people naturally connect with each other and learn.

Adapting and growing over time: Our platform is not static. Users are able to help shape it(for example, they can change the room themes ,they can create new buildings or even update the content by sharing a new artifact). This flexibility matches Roque's(2020) view of malleable computing as tools can grow and change with the users who use them.

Enabling Diverse Explorations in Time and Space

The design of our platform embraces all the dimensions of collaborative interaction through its flexible time-space framework.

Same time/Same place: Our platform allows users to meet in virtual spaces using live videochats, with avatars that move around and interact with each other like in real life. This type of interaction makes online gatherings feel more natural and lively, just like being in the same room. It is perfect for activities where people learn or create together, like in a cooking class where you can watch and ask questions in real time or the repair workshops where an expert can guide you through step by step. This design encourages easy conversations just like face-to-face hangouts.

Same time/Different place: In our platform distance is not an obstacle because people from anywhere can connect in real time for live videochats and collaboration. The technology keeps conversations smooth and lag free, ensuring that conversations flow naturally despite participants being continents apart. For example,

you can practice languages with native speakers across the world and also you can do a quick skill sharing with people you would never meet otherwise. Because all of these interactions happen instantly, these meetups become more lively.

Different time/Same place: Every room maintains a collection of artifacts that stay permanently available as community resources. Over time, these spaces become deep knowledge centers around specific topics where ideas are built on each other. Users can return to old artifacts, add new thoughts-comments or even combine different artifacts to create deeper understandings. This persistent layer turns each room into a living repository where artifacts-knowledge does not disappear after the live videochat ends, but instead it becomes a starting point for future visitors to work with and build on.

Different time/Different place: The archive allows people to access shared knowledge anytime and from anywhere, working like these digital tools that help teams collaborate without the need of being online at the same time.

This approach ensures that the platform adapts to how people actually work together. Sometimes synchronously, sometimes apart but always connected through shared purpose and knowledge. The platform supports what Bardram identifies as the dynamic needs of cooperative work.

Technical Modules: Graphs, Recommender Systems and Shared Objects

Some of the technical modules discussed in class are part of the platform:

Graphs: The social network shows user connections which can be shown as a graph where nodes represent users and edges represent relationships. By analyzing this graph (for example counting how many connections each user has), we can find some important people in the community. These important users might get some special permissions, such as the ability to create their own building.

Recommender System: Even though, our prototype does not have a recommendation system, it uses some basic ideas from collaborative filtering. For example: Friend suggestions could be based on users who interact with similar things-similar to how systems like GroupLens work. Also, popular artifacts (the ones which have the more "thank yous" could get shown more prominently.

Shared Online Objects: The artifacts in our platform act like flexible digital objects which can be changed and reused at any time. This concept comes from the researches by Roque(2020) and Bodker, who studied about how such objects help

collaboration. These artifacts stay available and can be modified by the users of the platform. Also, the artifacts work as “boundary objects” which means that different groups can use and interpret them in their own way while still sharing a same resource. This aligns with Bannon and Bodker’s concept of systems that are open and adaptable for collaboration.

Supporting Malleability and Appropriation

The platform incorporates many features that allow for continuous adaptation and change. Key features include:

User-Generated spaces: Users who have gained high-reputation are able to create new buildings, enabling what Shove et al. (2012) would call as practice innovation through material configurations. This allows and encourages creative and new ways of using the platform’s possibilities.

Artifact Customization: Users are able to leave and share various types of media-artifacts(text, images ,etc.) ,supporting what Roque(2020) and Bodker identify as different forms of appropriation. We have re-materialization (combining artifacts), re-mediation (using artifacts in new contexts) and re-interpretation(finding new meanings in the existing artifacts).

Adjustable layouts: The admins are able to reorganize the virtual city’s layout, creating what Roque(2020) and Bodker refer to as “interoperable reconfigurations” that allow new uses of the environment over time.

Unlike the rigid digital systems, this design allows the platform to grow with its users, which is the main idea in the practice-centered computing.

Future Development and Technical Limitations

The current prototype of Tell Me Something demonstrates the main ideas, but there are some key features that still need to be developed. These will help make the project more interactive and knowledge sharing, as it was originally intended.

It is needed to fully develop the Artifact and Billboard System. Right now, we are using Miro Boards, but for the final version we want to replace it with a real system for uploading ,organizing and filtering content. I would also be good to have a way to recommend useful artifacts and let users leave a “Thank you” to show appreciation to others people contributions.

Furthermore, an archive system needs to be created in order to keep knowledge alive over time and automatically save old or unused content. It would also be helpful to create a special “Archive Building” where users can still access past contributions to preserve shared contributions, reflecting Bannon and Bodker’s ideas on organizational memory.

Some important features like events , reputaion-based permissions (for creating houses) and moderation tools (like voting to remove users or for temporary bans) were only partly built in the prototype. All of these are needed to let the community govern itself in a fairly way.

Finally, to support malleability and appropriation, trusted users should be allowed to change the layout of the virtual city, so it can grow and adapt as a community’s needs change. This matches the idea that digital environments should evolve with their users.

Conclusion: Bridging Theory and Practice

“Tell Me Something” tackles away the problems of digital life like those that Byung-Chu Han talks about, while making a space for real human connections. In contrast with the typical social media, we built this platform using theories about teamwork, shared culture information and everyday habits ,so it can actually help users-people connect with each other in a deeper way.

Our platform’s design makes digital spaces and interactions feel more real and tangible which fixes what Han calls the emptiness of “non-things”. At the same time, the artifact system creates what Bush recognize as “trails” through collective knowledge. The reputation system help and encourages the community to work together purposefully while flexible spaces let users adapt and reshape the platform’s environment to their needs.

Some future applications could extend to professional communities of practice(for example, work teams-sharing skills and knowledge), educational environments(for example, classrooms-making learning more hands-on, online learning) or cultural preservation initiatives (for example, cultural projects-preserving traditions digitally). “Tell Me Something” is a place for real,meaningful, human-centered connections through shared activities.

We have not just created another digital platform, we created a place where knowledge stays connected to real human experiences, people can enjoy simply being together and build deep connections through shared activities and goals.

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